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	Interface and abstract class	Aparichaa 9/12/25

Lab Program-01

① Display Fibonacci series up to n terms using command line arguments

```
class Fib {
    public static void main(String[] args) {
        if (args.length != 1)
        {
            System.out.println("Dont take
    arguments more than one");
        return;
        }
    int n = Integer.parseInt(args[0]);
    int first = 0, second = 1;
    System.out.println("Fibonacci series up to"
    + n + "terms :");
    for(int i = 1; i <= n; i++) {
    System.out.println(first);
    int next = first + second;
     first = second;
     second = next;
     }
     }
     }
```

Output

Fibonacci series up to 5 terms

0
1
1
2
3
5

Lab Program - 02

② Demonstrate single Inheritance.

```
// Parent class
class Animal {
    void eat() {
    System.out.println("Animals can eat.");
    }
}

//child class (inherits from Animal)
class Dog extends Animal {
    void bark() {
    System.out.println("Dog can bark:");
    }
}

public class SingleInheritanceDemo{
    public static void main (String[] args){
        Dog d = new Dog();
    // using method from parent class
        d.eat();
    // using method from child class
        d.bark();
    }
}
```

Lab Program -03

③ Sort n elements using an array

```
import java.util.Scanner;
public class SortArray{
    public static void main (String[] args) {
     Scanner sc = new Scanner(System.in);
System.out.print("Enter number of elements:");
int n= sc.nextInt();
int[] arr = new int[n];
System.out.println("Enter "+n+"elements:");
for (int i=0; i<n;i++) {
arr[i] = sc.nextInt();
 }
for (int i= 0; i<n-1; i++){
for (int j=0; j<n-1-i; j++) {
if (arr[j] >arr[j+1]) {
//swap
int temp = arr[j];
arr[j] = arr[j+1];
arr[j+1] = temp;
    }
    }
  }
```

```
System.out.println("Sorted Elements:");
for (int i = 0; i<n; i++){
System.out.println(arr[i]);
    }
 sc.close();
  }
```

Output

```
Enter number of elements  5
Enter 5 elements:
  10
   5
  200
   8
   1.
Sorted elements: 1
                 5
                 8
                 10
                 200
```

Lab Program-04

④ Implement constructor overloading by passing different number of parametrs of different typy

```
class Student {
 String name;
 int age;
 double marks;
 //constructor with no parameters
 Student (){
  name = "Unknown";
  age = 0;
  marks = 0.0;
 }
 //constructor with one parametr (name only)
 Student (String n) {
  name = n;
  age = 0;
  marks = 0.0;
 }
 //constructor with two parametrs (name, age)
 Student (String n, int a) {
  name = n;
```

```
        age = a;
        marks = 0.0;
    }

    // Constructor with three parameters (name, age, marks)
    Student(String n, int a, double m) {
        name = n;
        age = a;
        marks = m;
    }

    void display() {
        System.out.println("Name: " + name + ", Age: "
        + age + ", Marks: " + marks);
    }
}

public class ConstructorOverloadingDemo {
    public static void main(String[] args) {
        // Using different constructors
        Student s1 = new Student();
        Student s2 = new Student("Alice");
        Student s3 = new Student("Bob", 20);
        Student s4 = new Student("Charlie",
                                 22, 85.5);
```

```
// Display Details
S1.display();
S2.display();
S3.display();
S4.display();
}
}
```

Output

```
Name: Unknown, Age: 0, Marks: 0.0
Name: Alice, Age: 0, Marks: 0.0
Name: Bob, Age: 20, Marks: 0.0
Name: Charlie, Age: 22, Marks: 85.5
```

Lab Program-05

⑤ Demonstrate string Methods

```
public class Length {
  public static void main (String[] args) {
    String str1 = "Hello World";
    String str2 = "Java Programming";

  // length()
  System.out.println("Length of str1: "+str1.length());

  // charAt()
  System.out.println("Character at index 4: "+
                     str1.charAt(4));

  // toUpperCase() and toLowerCase()
  System.out.println("Upper case: "+str1.toUpperCase
                     ());
  System.out.println("Lowercase: "+str1.toLowerCase
                     ());

  //trim()
  System.out.println("before trim: "+str2+" ");
  System.out.println("After trim: "+str2.trim()+
                     " ");

  // substring()
  System.out.println("Substring (0,5): "+
```

```
    Str1.substring(0,5));

//equals() and equalsIgnoreCase()
System.out.println("Str1.equals 'Hello world'?"
                    + str1.equals("Hello World"));
System.out.println("str1 equalsIgnoreCase 'hello
 world'?" + str1.equalsIgnoreCase("hello world"));

//contains()
System.out.println("Does str1 contain 'world'?"
 + str1.contains("world"));

// replace()
System.out.println("Replace 'word' with 'Java':"
+ str1.contains("world","Java"));

// Split()
String[] words = str1.split(" ");
System.out.println("splitting str1 :");
 for (String word:words){
 System.out.println(word);
  }
  }
 }
```

Output

```
Length of str1: 11
Character at index 4: O
Uppercase: HELLO WORLD
Lowercase: hello world
Before trim: ' Java Programming'
After trim: 'Java Programming'
Substring (0,5): Hello
Str1 equals 'Hello World'? true
str1 equalsIgnoreCase 'hello world'? true
Does str1 contain 'World'? true
Replace 'World' with 'Java': Hello Java
Splitting str1:
Hello
World.
```

Lab Program - 06

⑥ Demonstrate Vector Methods

```
import java.util.*;
class VectorDemo{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Vector<String> v = new Vector<>();
        System.out.print("How many fruits? ");
        int n = sc.nextInt();
        sc.nextLine(); // to skip newline
        for(int i=0; i<n; i++){
            System.out.print("Enter fruit name: ");
            String fruit = sc.nextLine();
            v.add(fruit);
        }
        System.out.println("\nFruits : " + v);
        System.out.print("enter a fruit to remove: ");
        String f = sc.nextLine();
        v.remove(f);
        System.out.println("After removing: " + v);
        sc.close();
    }
}
```

Output.

```
How many fruits?
3
Enter fruit name: apple
Enter fruit name : banana
Enter fruit name : orange.
Fruits : [apple, banana, orange]
Enter a fruit to remove: apple
After removing : [banana, orange]
```

Lab Program -07

⑦ Demonstrate concept of interface

```
// Interface
interface Animal {
    void eat();     //abstract method
    void sleep();   //abstract method
}
//class implementing interface
class Dog implements Animal{
    @Override
    public void eat(){
System.out.println("Dog eats bones.");
    }
    @Override
    public void sleep(){
    System.out.println("Dog sleeps in kennel.");
    }
}
// Another class implementing same interface
class Cat implements Animal {
    @Override
    public void eat(){
    System.out.println("Cat eats fish.");
    }
    @Override
    public void sleep() {
```

```
System.out.println("cat sleeps on sofa");
  }
}

public class InterfaceDemo{
    public static void main (String[] args){
Animal a1 = new Dog(); // ~~Inte~~ reference of
                          Interface, object of
                          Dog
Animal a2 = new Cat(); // reference of interface
                          object of cat

a1.eat();
a1.sleep();
a2.eat();
a2.sleep();
  }
}
```

Output

Dog eats bones
Dog sleeps in kennel
cat eats fish
cat sleeps on sofa

Lab Program - 08

(8) Demonstrate concept of creating, accessing and using a Package

```
//File : mypackage/Hello.java
package mypackage;    // This declares the package
public class Hello{
    public void sayHello(){
        System.out.println("Hello from mypackage!");
    }
}
```

```
// File : Main.java
import mypackage.Hello;    // import the package
public class Main{
    public static void main(String[] args){
        Hello obj = new Hello(); // create object of
                                 Hello class
        obj.sayHello();
    }
}
```

Output

Hello from mypackage!

Lab Program - 09

(9) Demonstrate multithreaded programming

```
//Method 1: Extending Thread class
class MyThread extends Thread {
public void run(){
  for(int i=1; i<=5;i++) {
System.out.println(" From myThread: "+i);
try{
Thread.sleep(500); // pause for 0.5 sec
} catch (InterruptedException e) {
  System.out.println(e);
}
}
}
}
//method2: Implementing Runnable Interface
class MyRunnable implements Runnable{
public void run(){
for(int i=1; i<=5;i++){
System.out.println(" From myrunnable: "+i);
try{
Thread.sleep(700); // pause for 0.7 sec
} catch (InterruptedException e) {
System.out.println(e);
```

```
        }
      }
    }
  }
public class Multithreading{
  public static void main (String[] args){
  // using thread class
MyThread t1 = new MyThread();

  //using Runnable Interface
  Thread t2 = new Thread (new MyRunable());

  // start both threads
    t1.start();
    t2. start();

// main thread work
  for(int i=1; i<=5;i++){
System.out.println(" From main Thread " + i );
  try {
Thread.sleep(600);  // pause for 0.6 sec
  }catch (InterruptedException e) {
System.out.println(e);
  }
  }
}
}
```

Output

From MyThread : 1
From Main Thread : 1
From MyRunnable : 1
From MyThread : 2
From MainThread : 2
From MyRunnable : 2
From My Thread : 2
From Main Thread : 2
From MyRunnable : 2
From MyThread : 3
From Main Thread : 4
From My Thread : 5
From MyRunnable : 4
From Main Thread : 5
From MyRunnable : 5

Lab Program -10

(10) Demonstrate thread priority

```
class MyThread extends java.lang.Thread
  {
  MyThread (String name){
    super (name);
    }

public  void run(){
   System.out.println(getName()+" is running with
   priority " + getPriority ());
   }
  }
public class ThreadPriorityDemo {
public static void main (String [] args){
MyThread t1 = new MyThread ("Thread -1");
MyThread t2 = new MyThread ("Thread -2");
MyThread t3 = new MyThread ("Thread -3");
t1.setPriority (java.lang.Thread.MIN_PRIORITY);
t2.setPriority(java.lang.Thread.NORM_PRIORITY);
t3.setPriority(java.lang.Thread.MAX_PRIORITY);
t1.start();
t2.start();
t3.start();
}}
```

Output

```
Thread-3 is running with priority 10
Thread-1 is running with priority 1
Thread-2 is running with priority 5
```

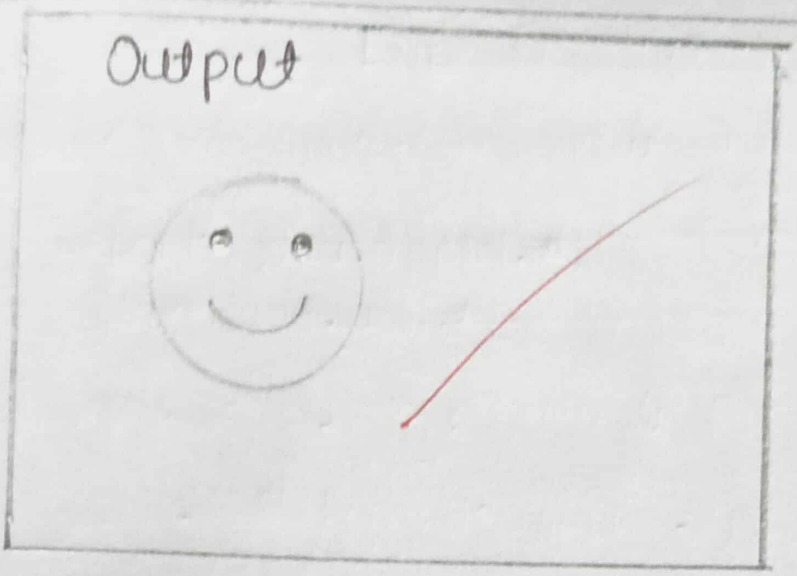
Lab Program - 11

(11) Create an applet to draw a human face.

```
import java.applet.*;
import java.awt.*;

public class face extends Applet{

public void paint(Graphics g){

g.setColor(Color.yellow);          //Face outline
g.fillOval(50,50,200,200);         //Face

g.setColor(color.black);
g.fillOval(100,100,20,20);         //Left eye
g.fillOval(180,100,20,20);         //Right eye

//Nose
g.setColor(Color.red);
g.drawArc(100,130,100,60,180,180);

}
}
```

```
<html>
<body>
<h3>Very Simple Human Face - Java Applet</h3>
<applet code="face.class" width="400" height="400">
</applet>
</body>
</html>
```

Output



Lab Program - 12

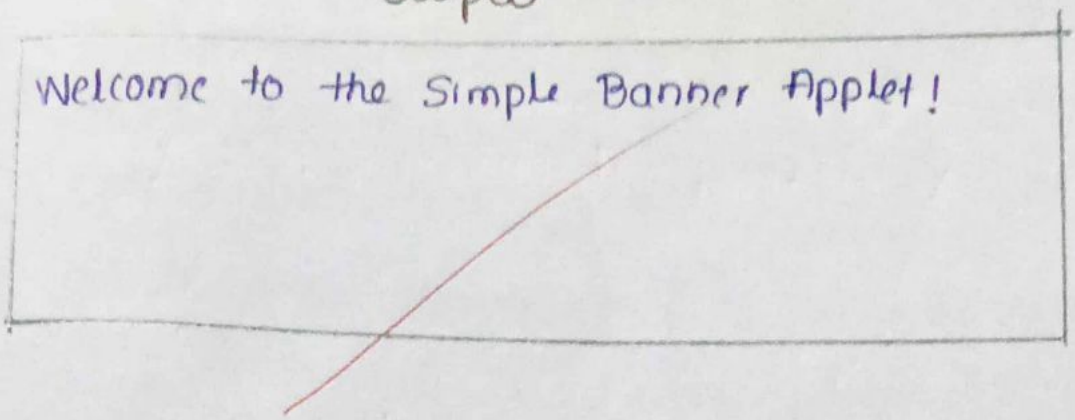
(12) Demonstrate simple banner applet.

```
import java.applet.Applet;
import java.awt.Graphics;

public class BannerApplet extends Applet implements
Runnable{
 String msg = "Welcome to the Simple Banner Applet!";
 Thread t;
 boolean stopFlag;

 public void init(){

  setBackground(Java.awt.Color.white);
  setForeground(Java.awt.Color.blue);

   }
 public void start(){
    t = new Thread(this);
   stopFlag = false;
   t.start();
    }
 public void run(){
  for(;;){
    try {
       repaint();
   Thread.sleep(150);
    msg. msg.substring(1)+msg.charAt(0);//to all
                                        characters
```

```
if (stopFlag)
  break;
 } catch (InterruptedException e) { }
  }
 }
public void stop(){
  stopFlag = true;
   t = null;
  }
public void paint(Graphics g) {
  g.drawString(msg, 20, 30);
  }
 }
```

```
<html>
<body>
  <applet code = "BannerApplet.class" width = "400"
   height = "60">
 </applet>
 </body>
 </html>
```

Output



Lab Program - 13

③ Program to count number of strings, integers and float values through command line arguments

```
public class countArgs {
    Public static void main(String[] args) {
     int countInt=0, countFloat=0, countString=0;
    for (String arg : args) {
    try {
    Integer.parseInt(arg);
    countInt++;
      } catch (NumberFormatException e1) {
    try {
      Float.parseInt(arg);
      countFloat++;
          } catch (NumberFormatException e2) {
      countString++;
        }
      }
    }
System.out.println ("Integers :" + countInt);
System.out.println (" Floats :" + countFloat);
System.out.println("Strings :" + countString);
```

}
}

Output

Chandana 123 , 50

Integer : 2

float : 0

string : 1.

Seen tutor

Lab Program - 14.

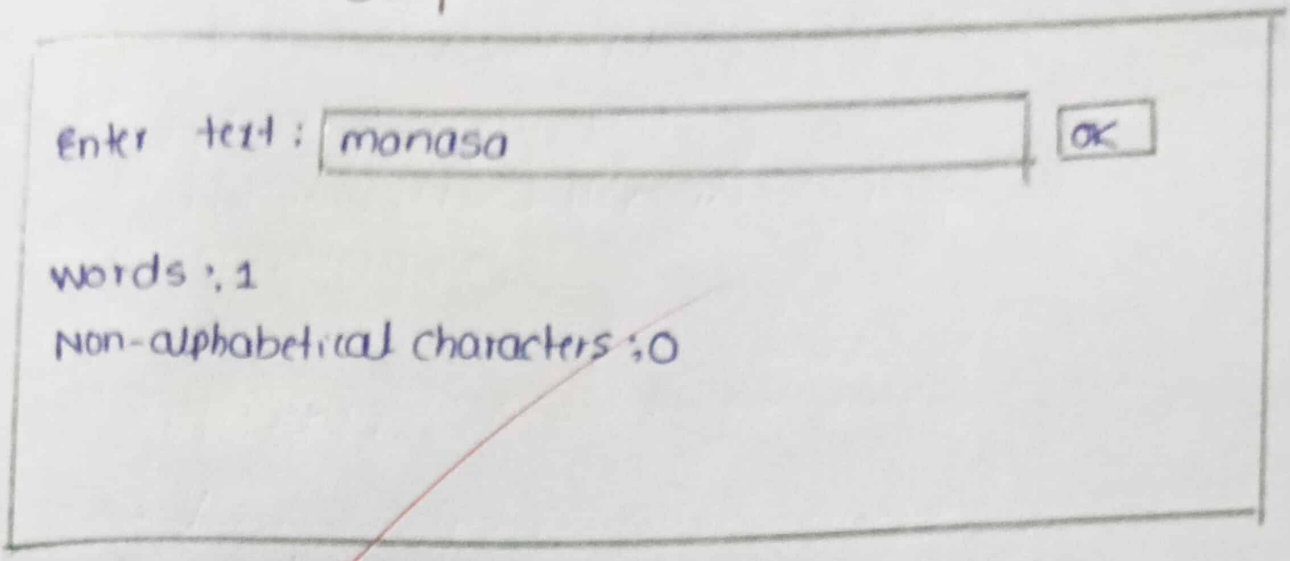
(14) Program to accept a message from the keyboard and display the no. of words and non-alphabetical characters

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class SimpleCountApplet extends Applet implements
ActionListener {

 TextField tf = new TextField(30);
 Button b = new Button("OK");
 int words = 0, nonAlpha = 0;

 public void init(){
     add(new Label("Enter text : "));
      add(tf);
      add(b);
       b.addActionListener(this);
      }

public void actionPerformed(ActionEvent e){
  String s = tf.getText();
   words = 0;
   nonAlpha = 0;

// Count words
  if (s.trim().length()>0 {
     words = s.trim().split("\\s+").length;
     }
```

```
//Counting non-alphabetical characters
for(int i=0; i<s.length(); i++){
    char ch = s.charAt(i);
    if(!Character.isLetter(ch) && ch != ' ')
     nonAlpha++;
    }
    repaint();
    }
public void paint(Graphics g){
    g.drawString("words: " + words, 20, 120);
    g.drawString("Non-alphabetical characters: " +
                    nonAlpha, 20, 140);
        }
    }
```

```
<html>
  <body>
    <applet code="SimpleCountApplet.class" width="400"
     height="200">
    </applet>
  </body>
</html>
```

Output



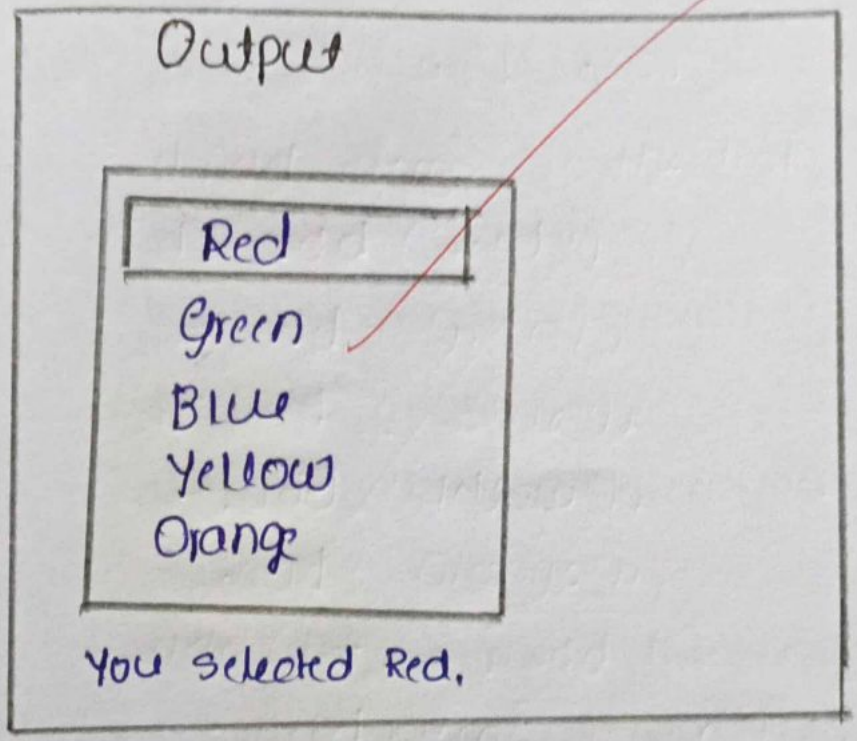
Lab Program -15

(15) Demonstrate creation of list using an applet.

```
import java.applet.Applet;
import java.awt.*;;
import java.awt.event.*;

public class list extends Applet implements ItemListener{
List lst;
String msg = " ";
public void init(){
    //Create a list with 5 visible rows (single-selection)
     lst = new List(5);
    // Add items to the list
     lst.add("Red");
    lst.add("Green");
    lst.add("Blue");
    lst.add("Yellow");
    lst.add("Orange");
    add(lst);  // add list to applet by one
    lst.addItemListener(this);
    }
public void itemStateChanged(ItemEvent e) {
  msg = "You selected: " + lst.getSelectedItem();
  repaint();
  }
public void paint(Graphics g){
```

```
g.drawString(msg, 20, 150);
  }
}
```

```
<html>
<body>
  <applet code = "list.class" width = "300" height = "200">
  </applet>
</body>
</html>
```



Lab Program-16

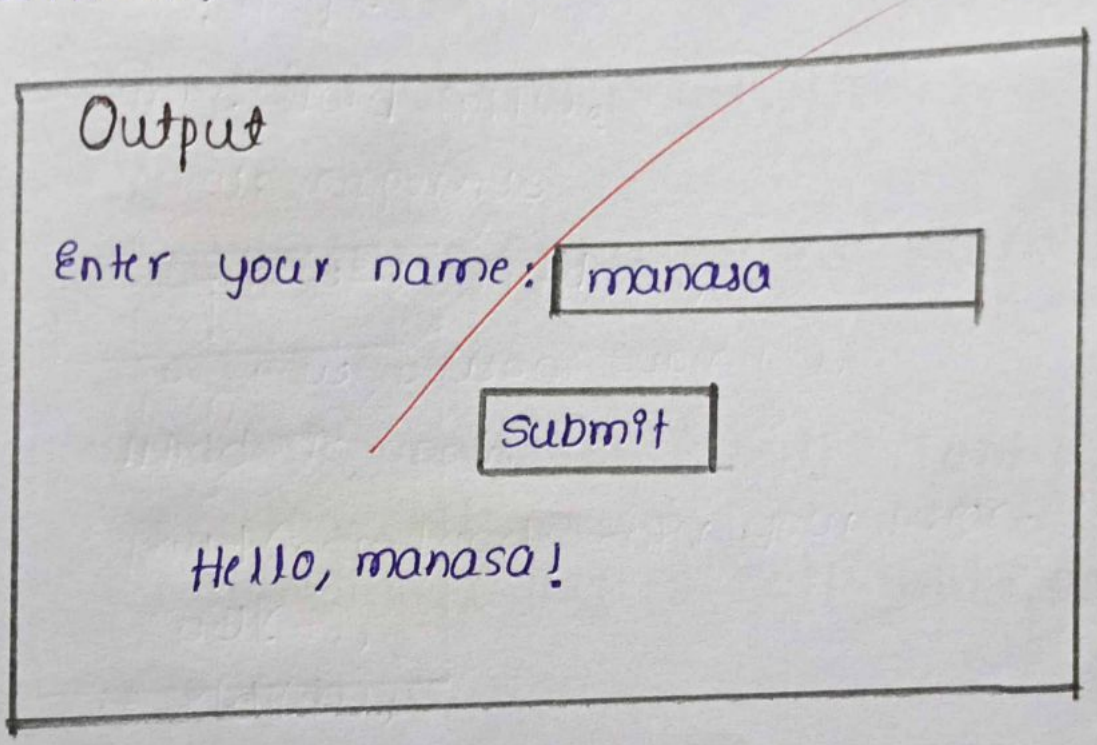
(16) Demonstrate concept of event handling

```
import java.applet.Applet.Applet;
import java.awt.*;
import java.awt.event.*;
public class simpleapplet extends Applet implements
ActionListener {
  Test
  TextField t;
  Button b;
  String msg = " ";
  public void init(){
    //create components
    t= new TextField(15);
    b = new Button("submit");
    //Add to applet
    add(new Label("Enter your name:"));
    add(t);
    add(b);
//Add action listener to button
  b.addActionListener(this);
  }
public void actionPerformed(ActionEvent e) {
  msg = "Hello," + t.getText() + "!";
  repaint();
  }
```

```
public void paint (Graphics g) {
    g.drawString (msg, 50, 100);
    }
    }
```

```
<html>
<body>
<applet code = "simpleapplet.class" width = "300"
 height = "150" > </applet >
</body>
</html>
```

Output



Lab Program - 17

(17) Program to demonstrate different types of fonts

```
import java.applet.Applet;
import java.awt.*;

public class font extends Applet{

    public class font extends Applet{

        public void paint(Graphics g){

    //Font : 1 : Plain

    Font f1 = new Font("Serif", Font.PLAIN, 20);
     g.setFont(f1);
     g.drawString("This is Serif plain", 20, 40);

    //Font 2: Bold

    Font f2 = new Font("Serif", Font.BOLD, 20);
     g.setFont(f2);
     g.drawString("This is Serif Bold", 20, 80);

    //Font 3: Italic

    Font f3 = new Font("Serif", Font.ITALIC, 20);
     g.setFont(f3);
     g.drawString("This is Serif Italic", 20, 120);

    //Font 4: Monospaced

    Font f4 = new Font("Monospaced", Font.BOLD, 22);
     g.setFont(f4);
     g.drawString("This is SansSerif", 20, 200);
```

```
}
}
```

```
<html>
<body>
  <applet code = "font.class" width = "400"
  height = "250" > </applet>
</body>
</html>
```

Output

This is Serif Plain

This is Serif Bold

This is Serif Italic

This is Monospaced

This is SansSerif

Lab Program -18

(18) Create an applet to tokenize the string

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;
import java.util.StringTokenizer;

public class TokenizeApplet extends Applet implements
ActionListener {

  TextFiud tf;
  Button b;
  String result = " ";

  public void init(){
     tf = new TextField(30);
     b = new Button("Tokenize");

  add(new Label("Enter a sentence:"));
  add(tf);
  add(b);
  b.addActionListener(this);
 }
public void actionPerformed(ActionEvent e){
 String input = tf.getText();
   result = " ";
//Tokenizing the String
StringTokenizer st = new StringTokenizer(input);
while(st.hasMoreTokens()){
```

```
result += st.nextToken() + "\n";
}
repaint();
}
public void paint(Graphics g){
    g.drawString("Tokens:",20,120);
    int y=140;
    for(String line: result.split("\n")){
      g.drawstring(line,20,y);
        y+=20;
      }
    }
}
```

```
<html>
<body>
    <applet code="TokenizerApplet.class" width="400"
    height="300"></applet>
</body>
</html>
```

Output

Enter a sentence: my name is sahana

Tokenize

Tokens:

my

name

is

sahana

Lab Program - 19

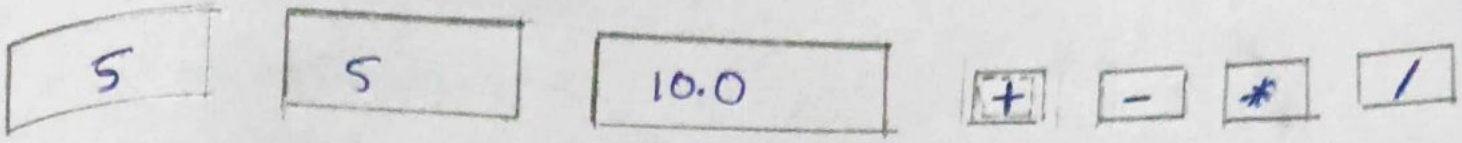
(19) Design a simple calculator using java applets

```
import java.applet.Applet;
import java.awt.*;
import java.awt.event.*;

public class calculator extends Applet implements
ActionListener{

TextField t1 = new TextField(5);
TextField t2 = new TextField(5);
TextField result = new TextField(10);

Button add = new Button("+");
Button sub = new Button("-");
Button mul = new Button("*");
Button div = new Button("/");

public void init(){
 add(t1);
 add(t2);
 add(result);
 add(add);
 add(sub);
 add(mul);
 add(div);
add.addActionListener(this);
sub.addActionListener(this);
mul.addActionListener(this);
div.addActionListener(this);
}
```

```
public void actionPerformed (ActionEvent e){
    double a = Double.parseDouble(t1.getText());
    double b = Double.parseDouble(t2.getText());
    double r = 0;

    if (e.getSource() == add) r = a+b;
    if (e.getSource() == sub) r = a-b;
    if (e.getSource() == mul) r = a*b;
    if (e.getSource() == div) r = a/b;

     result.setText("" +r);

     }
     }
```

```
<html>
<body>
<applet code = "calculator.class" width = "300"
 height = "150"> </applet>
</body>
</html>
```

Output



Lab Program -20.

(20) Implement static and dynamic stack using interface using abstract class

```
import java.applet.Applet;
import java.awt.Graphics;

//-- Stack Interface --
interface stack{
   void push(int x);
   int pop();
   }

//-- Abstract stack --
abstract class BaseStack implements stack{
 int arr[];
 int top = -1;

 BaseStack(int size){
   arr = new int[size];
   }
  }
//-- static stack (fixed size) --
class staticStack extends BaseStack{
staticStack (int size){
  super(size);
  }
public void push(int x){
    if (top == arr.length-1)
      return; // no space
```

```
arr[++top] = x;
}
public int pop() {
    if (top == -1)
      return -1; // empty
    return arr[top--];
    }
    }

//--- Dynamic Stack (grows automatically) ---
class DynamicStack extends BaseStack {
 DynamicStack(int size){
    super(size);
    }
 public void push(int x){
    if (top == arr.length-1){
     // grow
     int temp[] = new int[arr.length*2];
     for(int i=0; i<arr.length; i++)
      temp[i] = arr[i];
      arr = temp;
       }
    arr[++top] = x;
    }
    public int pop(){
       if(top == -1)
          return -1;
        return arr[top--];
```

```
}
}

//--- Applet ---
public class StackApplet extends Applet{
static Stack s;
DynamicStack d;
public void init(){
    s = new StaticStack(3);
    d = new DynamicStack(2);
    s.push(5);
    d.push(15);
    d.push(25); // grows
    }
public void paint(Graphics g){
    g.drawString("static pop:" + s.pop(), 20, 40);
    g.drawString("Dynamic pop:" + d.pop(), 20, 60);
    }
}
</html>
<body>
<applet code="StackApplet.class" width="300"
    height="150">
</applet>
</body>
</html>
```

Output

static pop: 30

Dynamic pop: 25

Seen
21/12/25